

ANA-ANDREEA STOICA

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Research Interests — *Machine Learning, Social Networks, Game Theory, Causal Inference.*

My research lies at the intersection of machine learning, statistics, economics, and policy, with a focus on understanding and designing AI systems in society. I combine tools from optimization, game theory, and behavioral and social modeling to study how AI systems interact with human preferences, incentives, and societal objectives.

ACADEMIC APPOINTMENTS

Research Group Leader, Max Planck Institute for Intelligent Systems, Tübingen, Germany <i>Social Foundations of Computation Department, led by Moritz Hardt</i>	2023-
Simons Fellow, Simons Institute For Theory of Computing University of California at Berkeley	Fall 2022
Doctoral Research Assistant, Computer Science Department, Columbia University <i>Advisor: Associate Professor Augustin Chaintreau</i>	2016-2022 New York, NY

EDUCATION

Columbia University, Ph.D, Computer Science Department M.Sc., M.Phil., Computer Science Department <i>Advisor: Associate Professor Augustin Chaintreau</i>	New York, NY 2022 2016, 2018
Princeton University, B.A., Mathematics Department Graduated <i>magna cum laude</i> . <i>Undergraduate Advisor: Professor Emmanuel Abbe</i> <i>Certificates: Applied & Computational Mathematics, Applications of Computing</i>	Princeton, NJ 2016

SELECTED FELLOWSHIPS, AWARDS, AND HONORS

P.I. for Alfred Sloan Foundation Grant 2022-19495 for ACM EAAMO'22, \$40,000	2022
Davide Giri Memorial Award for excellence in research and collaborations, Columbia University	2022
MIT EECS Rising Stars Workshop	2021
8th Heidelberg Laureate Forum	2021
J.P. Morgan AI Research PhD Fellowship	2019-2020
Andrew P. Kosoresow Memorial Award for excellence in teaching, Columbia University	2018
International Zhautykov Mathematics Olympiad, Bronze Medal	2010
Romanian National Mathematics Olympiad, 1 Gold, 2 Silver, and 3 Bronze medals	2007-2012
Diploma of Excellence awarded by Prime Minister of Romania	2008

PREPRINTS & OTHER WRITINGS

Haiwen Huang*, Vivian Nastl*, Tobias Holtdirk*, **Ana-Andreea Stoica**, Frauke Kreuter and Moritz Hardt. “A large-scale evaluation benchmark for social simulations.” *Manuscript in preparation*.

Tom Sühr, Nikhil Chandak, **Ana-Andreea Stoica*** and Moritz Hardt*. “An experimental platform for LLM evaluations: from pairwise comparisons to sequential content.” *Manuscript in preparation*.

Sierra Wyllie, Varun Gupta, **Ana-Andreea Stoica*** and Karolina Dziugate*. “Cents and sensitivity: data markets in machine learning.” *Manuscript in preparation*.

Ana-Andreea Stoica*, Michael Xu*, and Francesca Parise. “Committee mediated opinion dynamics in multi-community populations.” *Under submission*.

Faidra Monachou*, **Ana-Andreea Stoica***. “Fairness and equity in resource allocation and decision-making: an annotated reading list.” ACM SIGecom Exchanges, 2022.

PUBLICATIONS

Ana-Andreea Stoica, Celestine Mendler-Dünner, Moritz Hardt. “Stochastic wage suppression on gig platforms and how to organize against it.” Proceedings of the ACM Web Conference (WWW), 2026. [\[pdf\]](#)

- **EAAMO’26**: oral presentation at the ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization, November 2026.

Jiduan Wu, Rediet Abebe*, Moritz Hardt*, **Ana-Andreea Stoica***. “Policy Design in Long-run Welfare Dynamics.” Proceedings of the Thirteenth International Conference on Learning Representations (ICLR), 2025 [\[pdf\]](#)

- **EAAMO’25**: oral presentation at the ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization, November 2025.

Rashida Hakim*, **Ana-Andreea Stoica***, Christos H Papadimitriou, Mihalis Yannakakis. “The Fairness-Quality Trade-off in Clustering.” The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2024. [\[pdf\]](#)

- **LuWSI’25**: oral presentation at the 2nd Workshop on Learning Under Weakly Structured Information, March 2025.
- **ADT’24**: oral presentation at the 8th Intl. Conference on Algorithmic Decision Theory, October 2024.

Shubham Chowdhary, Giulia De Pasquale*, Nicolas Lanzetti*, **Ana-Andreea Stoica**, Florian Dörfler. “Fairness in Social Influence Maximization via Optimal Transport.” The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2024. [\[pdf\]](#)

- **EWAf’25**: oral presentation at the 4th European Workshop on Algorithmic Fairness, July 2025.

Ana-Andreea Stoica, Vivian Y. Nastl, and Moritz Hardt. “Causal Inference from Competing Treatments.” International Conference on Machine Learning (ICML), 2024. [\[pdf\]](#)

- **WALE’24**: oral presentation at the Workshop on Algorithms for Learning and Economics, June 2024.

Ana-Andreea Stoica, Nelly Litvak, and Augustin Chaintreau. “Fairness Rising from the Ranks: HITS and PageRank on Homophilic Networks.” Proceedings of the ACM Web Conference (WWW), 2024. [\[pdf\]](#)

- **ACM EAAMO’24**: oral presentation at the ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization, October 2024.

Ayse Gizem Yasar, Andrew Chong*, Evan Dong*, Thomas Gilbert*, Sarah Hladikova*, Carlos Mougan*, Xudong Shen*, Shubham Singh*, **Ana-Andreea Stoica***, and Savannah Thais*. “Integration of Generative AI in the Digital Markets Act: Contestability and Fairness from a Cross-Disciplinary Perspective.” LSE Legal Studies Working Paper, 2024. [\[pdf\]](#)

- **ML meets Law ’25**: oral presentation at Machine Learning meets Law, University of Tübingen, February 2025.
- **RegulatableML @ NeurIPS’24**: oral presentation at the RegulatableML Workshop at the Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), December 2024.
- **Article19’24**: oral presentation at the Article19 Symposium on DMA enforcement: Trends and gaps in the first year of application, November 2024.

Alexandra Olteanu et al. “FACTS-IR: Fairness, Accountability, Confidentiality, Transparency, and Safety in Information Retrieval.” In ACM SIGIR Forum, vol. 53, no. 2, pp. 20-43, 2021. [\[pdf\]](#)

Jessie Finocchiaro*, Roland Maio*, Faidra Monachou*, Gourab K Patro*, Manish Raghavan*, **Ana-Andreea Stoica***, and Stratis Tsirtsis*. “Bridging machine learning and mechanism design towards algorithmic fairness.” Proceedings of the Fairness, Accountability, and Transparency Conference (FAccT), 2021. [\[pdf\]](#)

- **AI4SG Workshop, IJCAI’20**: oral presentation at the AI for social good workshop at IJCAI, July 2020.

Ana-Andreea Stoica and Christos Papadimitriou. “Strategic Clustering.” Spotlight at NeurIPS’21 Workshop on Learning in Presence of Strategic Behavior. The Thirty-Fifth Annual Conference on Neural Information Processing Systems (NeurIPS), 2021.

Ana-Andreea Stoica, Abhijnan Chakraborty, Palash Dey, Krishna P. Gummadi. “Minimizing Margin of Victory for Fair Political and Educational Districting.” Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS). International Foundation for Autonomous Agents and Multiagent Systems, 2020. [\[pdf\]](#)

- **WINE’19**: poster at the Web and Internet Economics Conference, December 2019.

Ana-Andreea Stoica, Jessy Xinyi Han, Augustin Chaintreau. “Seeding Network Influence in Biased Networks and the Benefits of Diversity.” Proceedings of the ACM Web Conference (WWW), 2020. [\[pdf\]](#)

- **WINE’19**: poster at the Web and Internet Economics Conference, December 2019.
- **IC2S2’20**: poster at the International Conference on Computational Social Science, July 2020.

Ana-Andreea Stoica and Augustin Chaintreau. “Hegemony in Social Media and the effect of recommendations.” Companion Proceedings of the ACM Web Conference (WWW), 2019. [\[pdf\]](#)

Ana-Andreea Stoica and Augustin Chaintreau. “Fairness in the Social Influence Maximization Problem.” Companion Proceedings of the ACM Web Conference (WWW), 2019. [\[pdf\]](#)

Ana-Andreea Stoica, Christopher Riederer, and Augustin Chaintreau. “Algorithmic Glass Ceiling in Social Networks: The effects of social recommendations on network diversity.” Proceedings of the ACM Web Conference (WWW), 2018. [\[pdf\]](#)

- **MD4SG’18**: oral presentation at the Mechanism Design for Social Good Workshop, July 2018.
- **HALG’18**: oral presentation at the Highlights of Algorithms Workshop, June 2018.

Assimakis A. Kattis, Alexander Holiday, **Ana-Andreea Stoica**, and Ioannis G. Kevrekidis. “Modeling epidemics on adaptively evolving networks: A data-mining perspective.” Virulence 7.2 (2016): 153-162. [\[pdf\]](#)

**Equal contribution.*

Ana-Andreea Stoica. “Algorithmic Design for Social Networks: Inequality, Bias, and Diversity.” Columbia University Ph.D. Thesis, 2022. Advisor: Prof. Augustin Chaintreau. [\[pdf\]](#)

Ana-Andreea Stoica. “Community Detection in the Hypergraph Stochastic Block Model.” Princeton University Undergraduate Thesis, 2016. Advisor: Prof. Emmanuel Abbe. [\[pdf\]](#) Presented at:

- **New York Computer Science and Economics Day** (NYCE 2017) Poster session, New York University, New York, NY, USA. May 19th, 2017.
- **Fifth Networking Networking Women Workshop** (N² Women) Poster session, City University of New York, New York, NY, USA. October 3rd, 2016.

TALKS

Invited speaker, Dagstuhl Seminar on Rethinking Benchmarking, Germany	August, 2027
Invited keynote, 10th Anniversary Conference of ODISSEI, Utrecht, Netherlands	November, 2026
Invited organizer, Workshop on Algorithms for Learning and Economics, Greece	July, 2026
Invited speaker, Dagstuhl Seminar on Randomized Rounding, Germany	June, 2026
Invited speaker, Workshop on Social Choice and Learning, McGill University	May, 2026
Invited seminar, University of California at San Diego	April, 2026
Invited seminar, University of Toronto	April, 2026
Invited speaker, Hertie School of Business, Germany	April, 2026
Invited speaker, ELLIS Institute Finland	April, 2026
Invited seminar, Ecole Polytechnique, France	April, 2026
Invited seminar, Max Planck Institute for Software Systems, Germany	March, 2026
Invited seminar, University of Chicago	March, 2026
Invited seminar, Johns Hopkins University, Department of Applied Mathematics and Statistics	February, 2026
Invited seminar, University of Zürich	January, 2026
Invited seminar, Imperial College of London, Dyson School of Design Engineering	January, 2026
Invited tutorial, The Intl. Conference on Advances in Social Networks Analysis and Mining	August, 2025
Invited speaker, Cambridge Ellis Unit Summer School on Probabilistic Machine Learning	July, 2025
Invited speaker, Workshop on Algorithmic Fairness @ European Control Conference	June, 2025
Invited seminar, EconCS Seminar Series, Harvard University	March, 2025
Invited seminar, MIT	March, 2025
Invited speaker, 2nd Workshop on Learning Under Weakly Structured Information, U of Tübingen	March, 2025
Invited speaker, Machine Learning meets Law, AI & Law Institute, University of Tübingen	February, 2025
Invited seminar, Socially Intelligent Artificial Systems Group, University of Amsterdam	December, 2024
Invited seminar, Foundations of AI Seminar Series, University of Warwick	November, 2024
Invited speaker, PhilML'24 Conference, University of Tübingen	September, 2024
Invited seminar, Network Inequality Lecture Series, Complexity Science Hub	July, 2024
Invited speaker, Workshop on Humans, Algorithmic Decision-Making and Society, ICML	July, 2024
Invited speaker, Workshop on Algorithms for Learning and Economics, Greece	June, 2024
Invited seminar, Responsible AI Seminar Series, Archimedes AI Institute	April, 2024
Invited speaker, Machine Learning from Theory to Application Workshop, University of Tübingen	March, 2024
Invited speaker, Romanian Algorithms Day	July, 2023
Invited speaker, R U Social Conference, Reichman University	May, 2023
Invited seminar, Econometrics Seminar Series at the University of Oxford	April, 2023
Invited seminar, Psychology Department Seminar Series at New York University	September, 2022
Invited seminar, Data & Democracy Seminar, New York University	March, 2022
Invited seminar, Women in Combinatorics Colloquium Series	February, 2022
Invited seminar, Young Achievers Symposium, Pennsylvania State University	February, 2022
Invited seminar, Northeastern University and Network Science Institute	November, 2021
Invited seminar, Responsible Data Science, UIUC	November, 2021
Invited seminar, GRASP SFI Seminar Series, University of Pennsylvania	October, 2021

Invited seminar, University of Melbourne, Quantitative Methods Network	October, 2021
Invited speaker, INFORMS Annual Meeting, Diversity/PSOR/MIF	October, 2021
Invited speaker, QuantumBlack	August, 2021
Invited speaker, Romanian Algorithms Day	June, 2021
Invited speaker, Machine Learning Seminar Series, Morgan Stanley	May, 2021
Invited speaker, Computational Social Science Workshop, University of Chicago	April, 2021
Invited speaker, Harvard CRCS Rising Star in AI for Social Good, Harvard University	April, 2021
Invited poster, CAIS Symposium on Equity and AI, University of Southern California	April, 2021
Invited speaker, Theory of Computing for Fairness Seminar Series, Simons Institute	February, 2021
Invited speaker, ML Seminar Series, Oracle	February, 2021
Invited speaker, Fair February, Brown University	February, 2021
Invited panelist, NeurIPS 2020 Workshop on Fair AI in Finance	December, 2020
Invited speaker, The Analytical Foundations of Deep Learning Workshop, C3.ai Institute	October, 2020
Invited panelist, Inspired AI @ World Summit AI, virtual	October, 2020
Invited seminar, Rising Stars in Computer Science Seminar, UMass Amherst	September, 2020
Invited panelist, NIH Biomedical Informatics Coordinating Committee	August, 2020
Invited speaker, MGGG Graphs & Networks Workshop, Tufts University	July, 2020
Invited speaker, BitHacks hackathon, New York	June, 2020
Invited speaker, Impact Fellowship, New York	January, 2020
Invited panelist, Fordham IPLJ Symposium, Fordham University, New York, NY	October, 2019
Invited speaker, Women in Data Science Workshop at the Web Conference '18, Lyon, France	April, 2018
Invited seminar, Ecole Polytechnique Federale de Lausanne, Lausanne, Switzerland	May, 2018
Invited seminar, Cambridge University, Cambridge, UK	May, 2018
Invited seminar, Alan Turing Institute, London, UK	May, 2018

LEADERSHIP AND PROFESSIONAL SERVICE

ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (ACM EAAMO) [[website](#)] 2021–

- Co-founder, Program Co-Chair for ACM EAAMO'21, and Executive Committee member for the inaugural ACM EAAMO Conferences series.

EAAMO Bridges (formerly known as Mechanism Design for Social Good) [[website](#)] 2019–

- Director of Finance for the EAAMO non-profit organization, an interdisciplinary research initiative that includes workshops, colloquiums, and partnerships with academics, non-profits, and community organizations.

Area Chair & Program Committee

- Senior Program Committee: EC'26, AISI Track at AAAI'27
- Area Chair: Mechanism Design for Social Good Workshop 2020 (and Member of the Steering Committee) [[website](#)], ICLR 2026 Algorithmic Fairness Across Alignment Procedures and Agentic Systems (AFAA) Workshop [[website](#)]
- Program Committee Member: EWAFF'25-'26, EC'25, ACM FAccT'22-'25, NeurIPS'21,'24-'26, ECAI'24-'25, IC2S2'22-'25, EWAFF'23-'25, TheWebConf'19-'24, AISTATS'21-'24, AAAI'20-'23, FAccTRec'21-'23, WSDM'22, ACM COMPASS'21, FILA'20, Workshop on AI for Social Good @ IJCAI'19-'21, Workshop on AI for Social Good @ NeurIPS'19, Workshop on Fairness, Accountability, Confidentiality, Transparency, and Safety in IR @ SIGIR'19.
- Reviewer: Journal of Complex Networks, Journal of Causal Inference, Elsevier Fuzzy Sets and Systems Journal, Springer Nature Autonomous Agents and Multi-Agent Systems Journal, Journal of the Operations Research Society, IEEE/ACM Transactions on Networking Journal.

AI4ALL Summer Program

June-July 2020, 2021

- Led a team of instructors for the AI4ALL Summer Program for high school students, teaching courses on various topics in AI and mentoring students.

PROFESSIONAL EXPERIENCE

Research Intern

Summer, 2021

Advisor: Prof. Jon Kleinberg

Cornell University, NY

- Working at the intersection of algorithmic design and optimization, in understanding fairness and equity considerations in multi-stage optimization systems, with applications in public goods provisions.

Research Intern

Summer, 2019

Advisor: Prof. Krishna Gummadi

Max Planck Institute for Software Systems, Germany

- Designed voting mechanisms for clustering individuals into districts without creating disparate impact.

Research Intern

Spring, 2019

Advisor: Siddhartha Sen

Microsoft Research, NYC

- Worked at the intersection of machine learning and databases, developing machine learning algorithms for building a universal data index that is able to adapt to different workloads in an online fashion. *Manuscript in preparation.*

Graph Clustering Algorithms

2015–2016

Senior Thesis with Prof. Emmanuel Abbe

Princeton University, NJ

- Extended random graph algorithms for recovering communities in the general hypergraph stochastic block model.

Modeling epidemics on adaptively evolving networks

2015–2016

Independent Work with Prof. Yannis Kevrekidis

Princeton University, NJ

- Implemented dimensionality reduction techniques on random graph algorithms modeling epidemics.

TEACHING EXPERIENCE

Guest Lectures

- Invited guest lecture at Cornell Tech, “Networks and Markets” Spring 2024
- Invited guest lecture at Columbia University, “Introduction to Networks and Crowds” Spring 2020
- Invited guest lecture at Columbia University, “Introduction to Social Networks” Spring 2018
- Invited guest lecture at Columbia University, “AI Safety, Ethics, and Policy” Spring 2018

Introduction to Social Networks

Fall, 2016; Spring 2018

Columbia University, New York, NY

- Teaching Assistant. Held weekly office hours, graded homework assignments, and assisted with final projects.

STUDENT MENTORSHIP

Tom Sühr, Ph.D. student, Max Planck Institute for Intelligent Systems, **co-advisor** 2025–

Vivian Nastl, Ph.D. student, Max Planck Institute for Intelligent Systems 2026–

Jiduan Wu, Ph.D. student, Max Planck Institute for Intelligent Systems July–December 2025

Rashida Hakim, Ph.D. student, Columbia University July 2023–July 2024

Chaim Eisenbach, Undergraduate student, Columbia University January 2021–May 2021

Jessy Xinyi Wang, Undergraduate student, Columbia University June 2019–May 2020

MEDIA COVERAGE

Stochastic wage suppression on gig platforms and how to organize against it:

- Max Planck Institute for Intelligent Systems Press Release.

Algorithmic Glass Ceiling in Social Networks:

- Columbia Engineering News, la Repubblica, The Tribune, The Times of India, The Morning Paper.

The 1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization:

- AI Matters, Columbia Computer Science News, INFORMS Magazine.